

SYDNEY TECHNICAL HIGH SCHOOL



MATHEMATICS

Year 11

2 Unit

April 2007

Common Test

Time Allowed: 70 mins

Name :

Teacher :

Instructions:

- Begin each question on a new page
- Marks shown are approximate and may be varied
- Show necessary working
- Full marks may not be awarded if working is poorly set out or difficult to read
- Write all answers in simplest form

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	TOTAL
/8	/8	/8	/9	/8	/8	/8	/8	/65

Question 1

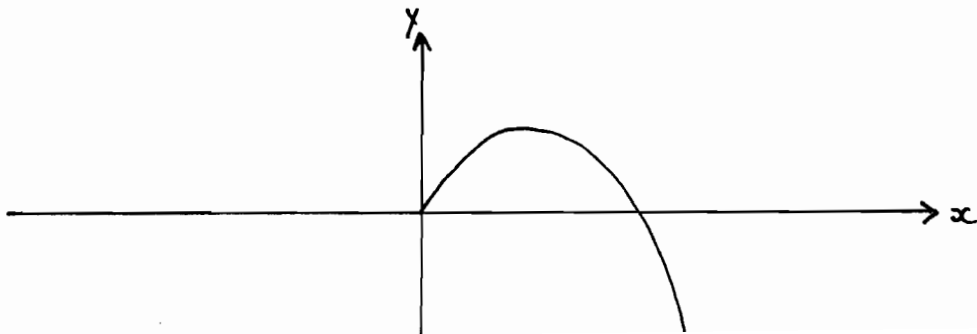
- a) Evaluate $\frac{2}{(1.11)^5 - 1}$ correct to 3 decimal places 2
- b) Express 68272 correct to 2 significant figures 1
- c) Write $\frac{x}{y} - 2$ as a single fraction 1
- d) Given that the surface area of an open cylinder is given by
 $A = 2\pi rh + \pi r^2$
find the value of h when $A = 250$ and $r = 6$ correct to 1 decimal place 2
- e) Subtract $3x^2 - 1$ from $x^3 - 2x^2 + 3$ 2

Question 2

- a) Find the exact value of $x^2 - 4x$ if $x = 4\sqrt{3}$ 2
- b) Rationalise the denominator of $\frac{\sqrt{3}}{6 - \sqrt{3}}$ 2
- c) Simplify $\sqrt{\frac{a^3 b^7}{ab^3}}$ 2
- d) State the range of the function $y = x^2 - 1$ 1
- e) The diagram shows part of a function $y = f(x)$ 1

Copy this diagram onto your answer sheet.

Complete the graph of $y = f(x)$ given that it is an odd function.



Question 3

a) Factorise fully

(i) $x^4 - x^2$ 2

(ii) $3a^2 + 2a - 8$ 2

(iii) $xy + 6x - y - 6$ 2

b) Simplify $\frac{a^3 + 1}{a^2 - a + 1}$ 2

Question 4

a) Solve the following

(i) $\frac{x-5}{4} + 3 = \frac{5x}{3}$ 2

(ii) $4x^2 - 4x + 1 = 0$ 2

(iii) $|x + 2| \leq 2$ 2

(iv) $|x - 1| = 3x - 1$ 3

Question 5

a) Sketch the following functions on separate number planes. Use a ruler to draw the axes. Label any important points.

(i) $y = \frac{2}{x}$ 1

(ii) $y = 3 - x$ 2

(iii) $y = 9 - x^2$ 2

b) Consider the function $y = \sqrt{16 - x^2}$

(i) Explain why the domain of this function is restricted 1

(ii) Find this domain 1

(iii) Sketch the function on a number plane 1

Question 6

- a) (i) Solve simultaneously:

2

$$2x + 3y = 4$$

$$3x + 6y = 2$$

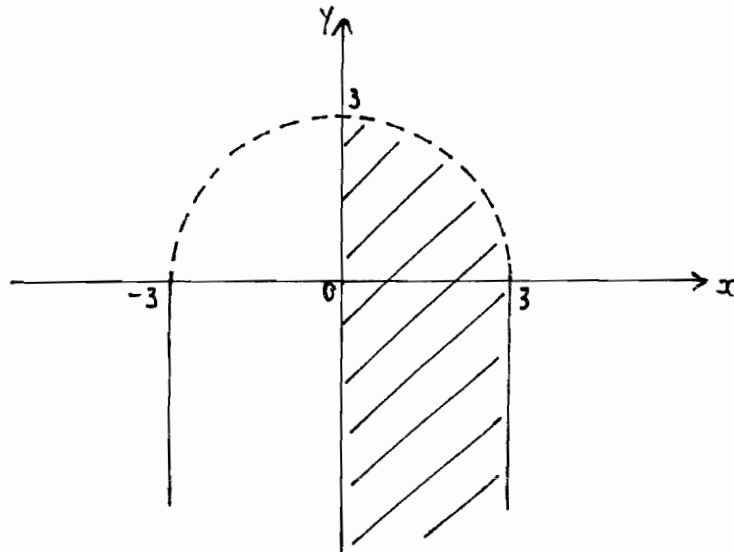
- (ii) What does this solution represent in relation to the graphs of

$$2x + 3y = 4 \text{ and } 3x + 6y = 2 ?$$

1

- b) State the two inequalities which represent the shaded region below:

2



- c) Express $0.\dot{3}\dot{6}$ as a simple fraction.

3

Question 7

- a) Find the values of a and b if $a + \sqrt{b} = 5 + 3\sqrt{6}$

2

- b) Simplify fully $\sqrt{98} \times \sqrt{48}$

2

- c) Solve $(x - 2)^2 = 3$ leaving your answer as a surd.

2

- d) The function $f(x)$ is defined as

2

$$f(x) = \begin{cases} x^2 + 1 & \text{if } x > 3 \\ 3x & \text{if } -2 \leq x \leq 3 \\ 2 & \text{if } x < -2 \end{cases}$$

Find $f(-3) + f(4) - f(3)$

Question 8

a) Simplify $\frac{2}{x-y} + \frac{y}{x^2-y^2}$ 2

b) Simplify $f(x) + f(-x)$ if $f(x)$ is an odd function 1

c) Determine if the following function is odd, even or neither. 2

Show all working. $f(x) = \frac{x^2+1}{x^2-1}$

d) (i) Expand $(x - \frac{1}{x})^2$ 1

(ii) If $x - \frac{1}{x} = 3$, find the value of $x^2 + \frac{1}{x^2}$ without solving for x 2

a) $\frac{2(x+y)}{x^2-y^2} = \frac{2x+3y}{x^2-y^2}$

b) $f(-x) = \frac{(-x)^2+1}{(-x)^2-1} = \frac{x^2+1}{x^2-1} = f(x)$ ∴ even

c)

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Student's Name/N°:

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Question 1

a) $\frac{2.919}{\text{---}}$

(2)

b) 68000 or

$\frac{6.8 \times 10^4}{\text{---}}$

(1)

c) $\frac{x}{y} - 2$

$\frac{xc}{y} - \frac{2y}{y}$

$= \frac{x - 2y}{y}$ (1)

d) $A = 2\pi rh + \pi r^2$

$250 = 2\pi \times 6 \times h + \pi \times 6^2$ (1)

$h = 3.6$ (1)

e) $x^3 - 2x^2 + 3 - (3x^2 - 1)$ (1)

$x^3 - 5x^2 + 4$ (1)

Question 2

a) $x^2 - 4x$

$(4\sqrt{3})^2 - 4(4\sqrt{3})$ (1)

$16 \times 3 - 16\sqrt{3}$

$48 - 16\sqrt{3}$ (1)

b) $\frac{\sqrt{3}}{6 - \sqrt{3}} \times \frac{6 + \sqrt{3}}{6 + \sqrt{3}}$

$\frac{6\sqrt{3} + 3}{\text{---}}$

$\frac{33}{\text{---}}$

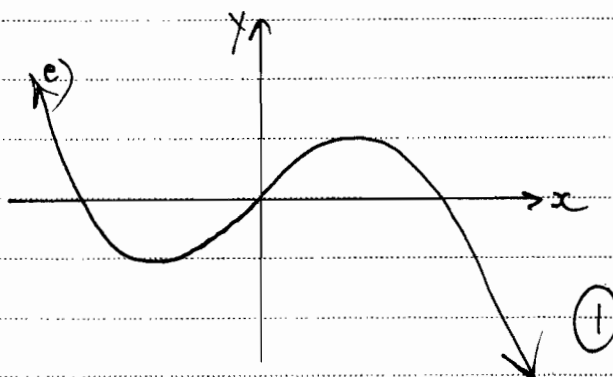
$\frac{2\sqrt{3} + 1}{11}$ (1)

c) $\frac{a^3 b^7}{\sqrt{a b^3}}$

$\sqrt{a^2 b^4}$ (1)

$= ab^2$ (1)

d) $y = x^2 - 1$

Range: All real $y \geq -1$ (1)Question 3

a) i) $x^4 - x^2$

$x^2(x^2 - 1)$ (1)

$x^2(x - 1)(x + 1)$ (1)

ii) $3a^2 + 2a - 8$

$(3a - 4)(a + 2)$

(2)

iii) $xy + 6x - y - 6$

$x(y + 6) - 1(y + 6)$ (1)

$(x - 1)(y + 6)$ (1)

b) $a^3 + 1$

$= (a + 1)(a^2 - a + 1)$ (1)

$= (a + 1)$ (1)

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Question 4

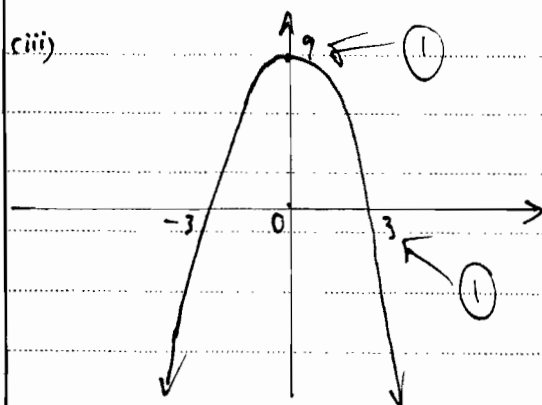
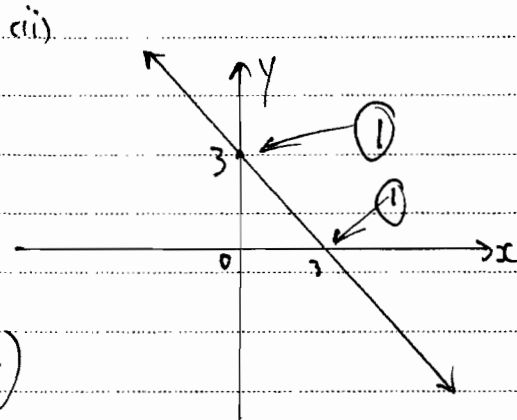
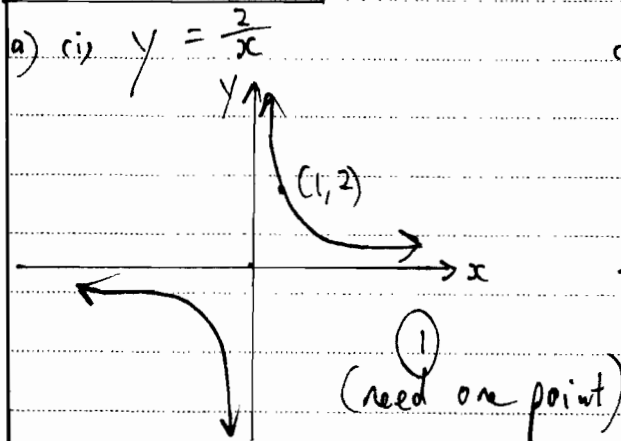
a) $\frac{x-5}{4} + 3 = \frac{5x}{3}$
 (i) $3(x-5) + 36 = 20x$
 $3x - 15 + 36 = 20x$ ①
 $21 = 17x$
 $x = \frac{21}{17}$ ①

(ii) $4x^2 - 4x + 1 = 0$
 $(2x-1)^2 = 0$ ①
 $x = \frac{1}{2}$ ①

(iii) $|x+2| \leq 2$
 $x+2 \leq 2$ or $x+2 \geq -2$ ①
 $x \leq 0$ or $x \geq -4$

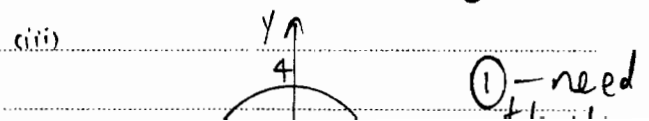
 $-4 \leq x \leq 0$ ①

(iv) $|x-1| = 3x-1$
 $x-1 = 3x-1$ or $x-1 = -(3x-1)$
 $0 = 2x$ or $x-1 = -3x+1$
 $x = 0$ or $4x = 2$
 $x = \frac{1}{2}$ ①
 Check! $x = \frac{1}{2}$ only ①

Question 5

b) (i) You can't $\sqrt{\text{a negative}}$ ①

(ii) $16 - x^2 \geq 0$
 $(4-x)(4+x) \geq 0$
 $-4 \leq x \leq 4$ ①



Teacher's Name: _____

Student's Name/N^o: _____Question 6

$$\begin{aligned} \text{a) (i)} \quad 2x + 3y &= 4 \quad \textcircled{1} \\ 3x + 6y &= 2 \quad \textcircled{2} \end{aligned}$$

$$\begin{aligned} 2 \times \textcircled{1} - \textcircled{2} \\ 4x + 6y &= 8 - \\ 3x + 6y &= 2 \\ \hline x &= 6 \end{aligned}$$

$$\begin{aligned} \therefore 2 \times 6 + 3y &= 4 \\ 3y &= -8 \end{aligned}$$

$$\underline{x = 6, \quad y = -\frac{8}{3}} \quad \textcircled{1} \quad \textcircled{1}$$

(ii) Point of intersection $\textcircled{1}$

$$\begin{aligned} \text{b) } x &\geq 0 \quad \text{and} \quad \textcircled{1} \\ y &< \sqrt{9-x^2} \\ \hline &\textcircled{1} \end{aligned}$$

$$\begin{aligned} \text{c) Let } x &= 0.3666 \dots \\ \therefore 10x &= 3.6666 \dots \quad \textcircled{1} \end{aligned}$$

$$\begin{aligned} \therefore 9x &= 3.3 \\ x &= \frac{3.3}{9} \\ &= \frac{33}{90} \quad \textcircled{1} \\ \underline{x} &= \underline{\frac{11}{30}} \quad \textcircled{1} \end{aligned}$$

Question 7

$$\begin{aligned} \text{a) } a + \sqrt{b} &= 5 + 3\sqrt{6} \\ a + \sqrt{b} &= 5 + \sqrt{54} \\ \therefore a &= 5, \quad b = 54 \\ \textcircled{1} \quad \textcircled{1} \end{aligned}$$

$$\begin{aligned} \text{b) } \sqrt{98} \times \sqrt{48} \\ &= 7\sqrt{2} \times 4\sqrt{3} \quad \textcircled{1} \\ &= 28\sqrt{6} \\ &\textcircled{1} \end{aligned}$$

$$\begin{aligned} \text{c) } (x-2)^2 &= 3 \\ x-2 &= \pm\sqrt{3} \\ \underline{x} &= \underline{2 \pm \sqrt{3}} \quad \textcircled{1} \end{aligned}$$

$$\begin{aligned} \text{d) } f(-3) + f(4) - f(3) \\ 2 + 4^2 + 1 - 3 \times 3 \quad \textcircled{1} \\ 2 + 17 - 9 \\ \underline{= 10} \\ \textcircled{1} \end{aligned}$$

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$$a) \frac{2}{x-y} + \frac{y}{x^2-y^2}$$

$$\frac{2}{x-y} + \frac{y}{(x-y)(x+y)}$$

$$\frac{2(x+y)}{(x-y)(x+y)} + \frac{y}{(x-y)(x+y)} \quad (1)$$

$$= \frac{2x+3y}{(x-y)(x+y)} \quad \text{or} \quad \frac{2x+3y}{x^2-y^2} \quad (1)$$

$$c) f(x) = \frac{x^2+1}{x^2-1}$$

$$\text{odd if } f(-x) = -f(x) \quad \text{even if } f(x) = f(-x) \quad (1)$$

$$\frac{(-x)^2+1}{(-x)^2-1} = -\left(\frac{x^2+1}{x^2-1}\right) \quad \frac{x^2+1}{x^2-1} = \frac{(-x)^2+1}{(-x)^2-1}$$

$$\frac{x^2+1}{x^2-1} \neq -\frac{x^2+1}{x^2-1}$$

 $\therefore$ Not odd

$$\frac{x^2+1}{x^2-1} = \frac{x^2+1}{x^2-1} \quad (1)$$

 $\therefore$ Even

$$d) \text{ i) } (x - \frac{1}{x})^2$$

$$= x^2 - 1 - 1 + \frac{1}{x^2}$$

$$= x^2 + \frac{1}{x^2} - 2$$

(1)

$$\text{ii) } (x - \frac{1}{x})^2 = x^2 + \frac{1}{x^2} - 2$$

$$\therefore x^2 + \frac{1}{x^2} = (x - \frac{1}{x})^2 + 2 \quad (1)$$

$$= 3^2 + 2$$

$$x^2 + \frac{1}{x^2} = 11$$

(1)